LAB # 11

NAÏVE BAYES

Objectives:

- To implement Naïve Bayes classifier in Python

Hardware/Software Required:

Hardware: Desktop/ Notebook Computer
Software Tool: Python 3.6/7/8

Introduction:

Naïve Bayes:

Naïve Bayes is one of the simplest probabilistic classifiers that classifies a candidate test vector based on Bayes Rule where all the features within the feature vector are assumed to be statistically independent:

$$P\left(w_i \mid x\right) = \frac{P\left(x \mid w_i\right) * P(w_i)}{P(x)}$$

where $P(x)$ is the posterior probability telling the conditional probability of the class w_i against the given feature vector x, $P\left(w_i\right)$is the likelihood of the feature vector x given the class w_i, $P(w_i)$ is the prior probability of class w_i and $P(x)$ is the evidence of feature vector x. Naïve Bayes can be used to classify both numerical and categorical test vectors. For classifying numerical test vectors, Naïve Bayes calculates likelihood probabilities of all features for each class labels through Gaussian probability density function. Then the likelihood probability $P\left(w_i\right)$ of a feature vector x is computed by taking the product of likelihood probabilities for all features. The Gaussian distribution can be computed through:

$$P\left(x \mid w_i\right) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where μ is the mean and σ is the standard deviation of the Gaussian distribution. The algorithm to implement Bayesian classification model is given below:

Algorithm: Naïve Bayes	
Input:	*Training data X*
	Training data labels Y

Sample x to classify
Output: *Decision y_p about sample x*

Compute prior probabilities 'p1' for the first-class
Compute prior probabilities 'p2' for the second-class
Compute likelihood for the test sample x against first-class training samples through Gaussian Distribution
Compute likelihood for the test sample x against second-class training samples through Gaussian Distribution

Compute posterior probability 'P1' by multiply all first-class likelihood probabilities for x with each other and then with p1.
Compute posterior probability 'P2' by multiply all second-class likelihood probabilities for x with each other and then with p2.

Compute evidence by summing P1 and P2.
Normalize P1 by dividing it with the evidence
Normalize P2 by dividing it with the evidence

If P1 >= P2

$$y_p = 0$$

else

$$y_p = 1$$

return y_p

Lab Task:

Task1:

Write a Python script to implement Naïve Bayes classification model for the following dataset and classify the given test sample:

Age	Loan	Class (Defaulter)
25	40000	0
35	60000	0
45	80000	0
20	20000	0
35	120000	0
52	18000	0
23	95000	1

40	62000	1
60	100000	1
48	220000	1
33	150000	1
48	**142000**	**?**

Task2:

Use the given diabetes dataset and classify it using Naïve Bayes classification model:

a) First create a Python script and load 'Diabeties.csv' file.
b) Identify features and classes from the loaded dataset.
c) Perform 2-fold cross validation on the dataset by splitting it into testing and training parts.
d) Implement a Naïve Bayes classifier using the above algorithm and use training dataset to classify each of the sample within testing dataset.
e) Compute the accuracy from the predicted test samples.

Conclusion:
Write the conclusion about this lab

NOTE: A lab journal is expected to be submitted for this lab.